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# ABS STRUCTURES TO ADDRESS CREDIT RISK

## Learning Outcome

- describe typical credit enhancement structures used in securitizations

A simple securitization transaction may involve the sale of only one class of ABS: Let us call this class Bond Class A. Returning to the Bright Wheel Automotive (BRWA) and Car Loan Trust (CLT) hypothetical example, CLT may raise EUR1,000 million by issuing 1,000,000 certificates for ABS Bond Class A with a par value of EUR1,000 per certificate. Thus, each certificate holder is entitled to 1/1,000,000 of the payments from the collateral, after payment of servicing and other administrative fees. In this case, the investors will be exposed directly to the default risk of the assets of the pool as well as other risks that impact the payments to the pool. There are different credit enhancement approaches the SPE can pursue to mitigate the impact of these risks on the investors.

These are achieved by creating securitization structures with multiple classes of ABS debt issued.

## Credit Enhancement

Most securitization structures involve multiple different tranches where each of the tranches plays a different role; often, one tranche assumes risk from other tranches. Credit risk is a core risk that securitization structures seek to mitigate through **credit enhancement**, a type of financial support that absorbs losses from defaults on the underlying loans. Three main types of **internal credit enhancements** are widely used in securitization transactions:

As introduced in Lesson 1 on covered bonds, overcollateralization involves a provision that a bond's collateral value exceeds the face value of the issued bonds. Even if there are defaults in the pool, the transaction has sufficient cushion to continue paying principal and interest payments on the bonds. Revisiting our previous example of the notes issued by Car Loan Trust, if the value of the loans and leases underlying the EUR1,000 million securitization is EUR1,200 million, there is EUR200 million overcollateralization. If the car loans perform poorly, let's say 10% default, there is still EUR1,080 million collateral to cover losses from other defaults. This additional cushion makes both the senior and subordinated tranches more attractive to potential investors.

**Excess spread** is the difference between the coupon on the underlying collateral and the coupon paid on the securities. For example, if the average loan in the CLT example generates 8% but the securities issued from the pool earn 6%, then the excess spread of 2% can absorb collateral shortfall or be used to build up reserves.

**Subordination** or **credit tranching** in the securitizationtransaction involves creating more than one bond class or tranche, and the bond classes differ in how they share any losses resulting from defaults in the collateral pool. The bond classes are classified as senior bond classes or subordinated bond classes; this structure is also referred to as a senior/subordinated structure. The subordinated bond classes are sometimes called "non-senior bond classes" or "junior bond classes." Since all investors in a securitization are paid from the same underlying cash flow pool, the tranching dictates the order of payments made to investors and the order in which losses are absorbed by the investors.

There are also **external credit enhancements**, such as financial guarantees by banks or insurance companies, letters of credit, and cash collateral accounts, which we do not cover here.

In November 2013, SolarCity, a predecessor organization of Tesla Energy, issued the first-ever US public solar ABS (SOCTY 2013-1) backed by USD54.425 million of bundled cash payments. This ABS allowed SolarCity to access a relatively cheap source of funding. Normal monthly payments by customers purchasing solar power from SolarCity generated the income stream used to pay ABS investors. The collateral comprised 5,033 photovoltaic (PV) systems located across 14 states, 143 counties, and 53 utility service territories. The bonds had a weighted average life of 7.05 years and a BBB+ credit rating from S&P, and they provided a 4.8% annual return to investors through 2026. The investment-grade rating meant pension funds and other regulated investment vehicles could own the ABS.

The ABS were overcollateralized versus the face value of the bonds, which meant that the ABS buyers received more collateral in case of default than the USD54.425 million purchase value. This overcollateralization was one of the reasons that the ABS received an investment-grade credit rating (BBB+ by S&P). The overcollateralization was also significant in facilitating the offering's 4.8% interest rate, which was a low cost of capital for solar developers. The 4.8% interest rate was only about 0.50% higher than the 30-year fixed-rate mortgage at that time. Lastly, the high amount of overcollateralization was required given that this was the first publicly known solar ABS, and investors needed to be comfortable in purchasing a new structured finance product.

SolarCity's 10-K for the fiscal year ending 31 December 2014 describes the assets backing these securities, as follows:

> *In November 2013, we pooled and transferred qualifying solar energy systems and the associated customer contracts into a special purpose entity, or SPE, and issued $54.4 million in aggregate principal of Solar Asset-backed Notes, Series 2013-1, backed by these solar assets to certain investors. The SPE is wholly owned by us and is consolidated in our financial statements. Accordingly, we did not recognize a gain or loss on the transfer of these solar assets. As of December 31, 2014, these solar assets had a carrying value of $143.9 million and are included under solar energy systems, leased and to be leased — net, in our consolidated balance sheets.*

Thus, collateral for these ABS was worth approximately 2.7 times the value of the ABS themselves, calculated as follows:

Total collateral USD143.9 million/Total value of ABS USD54.4 million

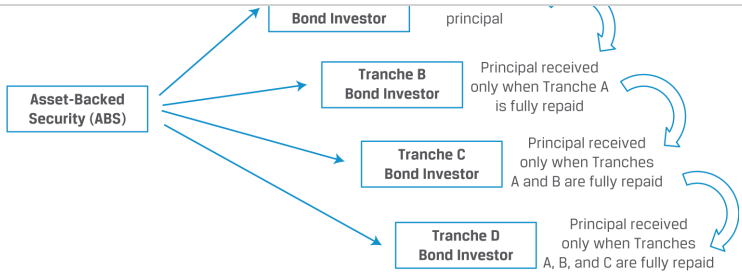
= 2.66 times.

Solar ABS securities such as this one can offer an attractive risk/reward profile for fixed-income investors. The securities are typically well-structured, with high levels of collateralization and credit enhancement, relatively low risk of default, and an attractive spread.

## Credit Tranching

Subordination functions as credit protection for the more senior bond classes; that is, losses are realized by the subordinated bond classes before any losses are realized by the senior bond classes. This type of protection is also commonly referred to as a "waterfall" structure because of the cascading flow of payments between bond classes in the event of default, as shown in Exhibit 3.

**Exhibit 3:**

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Let's revisit the BRWA securitization from an earlier module. In this case, CLT, the SPE, acquired the pool of car loans from BRWA. Assume that CLT then issues four bond classes with a total par value of EUR1,000 million: Bond Class A, the senior tranche, with a par value of EUR825 million; Bond Class B, the mezzanine tranche, with a par value of EUR100 million; Bond Class C, the junior tranche, with a par value of EUR50 million; and Bond Class D, a junior tranche that is subordinated to all the other bond classes and has a value of EUR25 million. In this structure, the four bond classes determine the capital structure of the securitization transaction.

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In the securitization structure shown in Exhibit 4, there is a prioritization of payments that determines the timing of interest and principal cash flows to the different tranches. The most senior investor tranche, with the highest ranking in the capital structure, Class A, receives principal payments first, carries the lowest risk, and offers the lowest return. The mezzanine tranche, Class B, assumes a slightly higher degree of risk but offers higher returns. Class D absorbs losses first and has the highest risk, with potentially the highest yield. The lowest tranche is sometimes referred to as the "equity" tranche because of the residual nature of its claims to cash flows from the asset pool.

The four tranches carry different yields and degrees of risk, and they are impacted differently in case of default. The specific terms and conditions are outlined in Exhibit 4. This structure is typical of securitizations.

### Exhibit 4:

### Overview of Notes Issued by Car Loan Trust – Brief Summary of Terms

| Four-Year Asset-Backed Notes Issued by CLT | | |
|---|---|---|
| **Issuer:** | Car Loan Trust (CLT) | |
| **Tranches:** | Tranche A Notes | |
| | Face value: | EUR825 million |
| | Interest rate: | MRR + 0.50% |
| | Credit enhancement features: | Subordination of the Class B Notes, the Class C Notes, and the Class D Notes |
| | Tranche B Notes | |
| | Face value: | EUR100 million |
| | Interest rate: | MRR + 1.50% |
| | Credit enhancement features: | Subordination of the Class C Notes and the Class D Notes |
| | Tranche C Notes | |
| | Face value: | EUR50 million |
| | Interest rate: | MRR + 2.50% |
| | Credit enhancement features: | Subordination of the Class D Notes |
| | Tranche D Notes | |
| | Face value: | EUR25 million |

features.

| | |
|---|---|
| **Status and Ranking of Payments:** | The Class A Notes are senior to the Class B, C, & D Notes.<br>The Class B Notes are senior to the Class C & D Notes and junior to the Class A Notes.<br>The Class C Notes are senior to the Class D Notes and junior to the Class A & B Notes.<br>The Class D Notes are junior to the Class A, B, & C Notes. |
| **Interest Payment:** | Monthly commencing 30 days from [Settlement Date] to be paid each month, with final payment on [Maturity Date] |
| **Seniority:** | The Notes are secured obligations of CLT |
| **Business Days:** | Frankfurt |



This senior/subordinated structure is an example of **credit tranching**. The most junior tranche, Class D, will absorb all losses up to EUR25 million. Should the credit losses exceed that threshold, Bond Class C will absorb all losses beyond the first EUR25 million, up to an additional EUR50 million. Consequently, Bond Class B will absorb losses beyond EUR75 million and up to an additional EUR100 million. Thus, if defaults in the CLT pool do not exceed EUR175 million, Bond Class A will be fully repaid its EUR825 million.

For example, if the total loss on the collateral in CLT is EUR70 million, then Class D Notes absorb the first EUR25 million of losses and are wiped out, and the Class D investors lose their investment. Then Bond Class C absorbs an additional EUR45 million of losses, and the Class C investors are repaid only the remaining EUR5 million. However, both Bond Class A and B continue to receive payments from the pool as agreed: They do not realize any loss in this scenario. Clearly, Bond Class A realizes a loss only if the total loss on the collateral exceeds EUR175 million and all the more junior tranches are wiped out.

This waterfall structure redistributes the credit risk associated with the collateral. The creation of a set of bond classes allows investors to choose the level of risk that they prefer to bear and receive a return accordingly. Moreover, by adjusting the features of the various tranches, investors can pick an appropriate combination of maturity, risk, and return characteristics.

## EXAMPLE 3

### Waterfall Calculation for ABS of Car Loan Trust

Similarly, if market reference rate (MRR) is 4.0%, an investor who purchased the following ABS issued by CLT would receive the following interest payments in the order shown.

| Investor's Initial CLT ABS Purchases | | Interest Paid If No Defaults in the CLT Pool | Interest Paid If Defaults in the CLT Pool Are EUR20 Million | Interest Paid If Defaults in the CLT Pool Are EUR90 Million |
|---|---|---|---|---|
| Class A Notes | EUR0 million | | | |
| Class B Notes | EUR3 million | EUR165 thousand | EUR165 thousand | EUR140 thousand |
| Class C Notes | EUR2 million | EUR130 thousand | EUR130 thousand | EUR0 thousand |
| Class D Notes | EUR0 million | | | |
| Total | EUR5 million | EUR295 thousand | EUR295 thousand | EUR140 thousand |

Risk ratings are assigned to each bond class created in the securitization, and these ratings depend on both the quality of the collateral and the seniority of the class. The risk ratings reflect the credit risk of the pool of securitized loans or receivables as well

C bonds. Depending on the structure of the securitization, each bond class in the transaction receives a risk rating that reflects its credit risk, and some of the bond classes may have a better credit rating than the company that is seeking to raise funds because of collateralization or credit enhancement.



## Question Set

**1.** *Determine the correct answers to complete the following sentences:*

It is common for securitizations to include a form of internal credit enhancement called _____, also referred to as _____. In such a structure, there is more than one bond class, and the bond classes differ as to how they will share any losses resulting from defaults of the borrowers whose loans are in the collateral.

Hide Solution

**Solution:**

It is common for securitizations to include a form of internal credit enhancement called subordination, also referred to as credit tranching. In such a structure, there is more than one bond class, and the bond classes differ as to how they will share any losses resulting from defaults of the borrowers whose loans are in the collateral.



For Questions 2 and 3, please refer to the additional information below regarding the ABS issued by Ahbaling Trust.

| Bond Class | Par Value (SGD millions) | Interest Rate | When Principal Is Repaid If No Prepayments in the Pool | Credit Enhancement Features |
|---|---|---|---|---|
| A (senior) | 280 | MRR + 1.0% | One year, per ABS terms | Subordination of the Class B and the Class C Notes |
| B (subordinated) | 60 | MRR + 2.0% | Two years, per ABS terms | Subordination of the Class C Notes |
| C (subordinated) | 60 | MRR + 4.0% | Three years, per ABS terms | None |
| Total | 400 | | | |

**2.** Calculate the value for each of the three bond classes *if* SGD140 million of the securitized receivables backing the ABS default.

Hide Solution

**Solution:**

| Bond Class | Initial Value (SGD millions) | After Default (SGD millions) |
|---|---|---|
| A (senior) | 280 | 260 |
| B (subordinated) | 60 | 0 |
| C (subordinated) | 60 | 0 |

The rules for the distribution of losses are as follows. All losses on the collateral are absorbed by Bond Class C before any losses are realized by Bond Class B and then Bond Class A. Consequently, if the losses on the collateral are

**3.** Identify the new maturity dates for the three bond classes if economic conditions worsen, driving up interest rates in year three.

Hide Solution

**Solution:**

| Bond Class | When Principal Is Repaid If No Prepayments in the ABS Pool | When Principal Is Repaid If Interest Rates Increase, Causing Some Borrowers to Slow Their Payments |
|---|---|---|
| A (senior) | One year, per ABS terms | One year, per ABS terms |
| B (subordinated) | Two years, per ABS terms | Two years, per ABS terms |
| C (subordinated) | Three years, per ABS terms | Potentially more than three years |

If an increase in interest rates results in financial distress for borrowers, some of them may slow down the speed of their payments. Since this takes place in year three, Class A and Class B bondholders already would have been repaid their principal. Only Class C bondholders would be impacted in this instance.

**4.** *Determine the correct answers to complete the following sentences:*

There are three main types of internal credit enhancements that are widely used in securitization transactions. _____ is a provision that the collateral underlying the transaction is larger than the face value of the issued bonds, providing a cushion for continued payments even with defaults in the pool. Excess spread is the difference between the coupon on the underlying collateral and the coupon paid on the securities, which can absorb collateral shortfall or build up reserves. With _____ or _____, the securitization transaction involves creating more than one bond class or tranche, and the bond classes differ in how they share any losses resulting from defaults in the collateral pool. The _____ determines the order of payments made to investors and the order losses are absorbed by the investors.

Hide Solution

**Solution:**

There are three main types of internal credit enhancements that are widely used in securitization transactions. Overcollateralization is a provision that the collateral underlying the transaction is larger than the face value of the issued bonds, providing a cushion for continued payments even with defaults in the pool. Excess spread is the difference between the coupon on the underlying collateral and the coupon paid on the securities, which can absorb collateral shortfall or build up reserves. With subordination or credit tranching, the securitization transaction involves creating more than one bond class or tranche, and the bond classes differ in how they share any losses resulting from defaults in the collateral pool. The tranching determines the order of payments made to investors, and the order losses are absorbed by the investors.

**5.** Which of the following statements about credit tranching is *incorrect*?

**A.** The waterfall structure redistributes the credit risk associated with the underlying collateral.
**B.** Some of the bond classes may have a better credit rating than the company that is seeking to raise funds.
**C.** Risk ratings are based on the market risk of the pool of securitized loans or receivables as well as the priority of how they absorb losses.

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risk) of the pool of securitized loans or receivables as well as the priority of how they absorb losses.

A is incorrect because it is a correct statement. The waterfall structure redistributes the credit risk associated with the collateral. The creation of a set of bond classes allows investors to choose the level of risk that they prefer to bear and receive return accordingly. B is incorrect because it is a correct statement. Each bond class in the transaction receives a risk rating that reflects its credit risk, and some of the bond classes may exhibit a better credit rating than the company that is seeking to raise funds because of collateralization or credit enhancement.

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